09. Interactive Charts with Plotly

Detailed beginner handbook - English and Arabic

Technologies: Plotly, react-plotly.js

What you will learn

Render a small interactive chart from synthetic totals.

Plotly creates browser charts with tooltips, zoom, selection, and image export.

الهدف: بناء مثال صغير وفهمه خطوة بخطوة.

شرح بسيط: تعلم الفكرة أولا، ثم جرب المثال الصغير ببيانات اصطناعية فقط.

How to study

Study one lesson at a time. Type examples yourself, predict the result, run the code, then change one thing. Keep a notebook of new words, errors, root causes, and fixes.

Course map

Setup and mental model; ten core concepts; a guided mini-project; the real dashboard architecture; debugging; exercises and answers.

Lesson 1 - Setup and mental model

Prerequisites

Windows 10 or 11, VS Code, PowerShell, and permission to install learning dependencies. Use only synthetic data in tutorials and screenshots.

Setup sequence

1. Open PowerShell.
2. Go to learning-examples/09-plotly.
3. Read README.md.
4. Run npm install then npm run dev.
5. Read the complete error if it fails.
6. Change one safe value and rerun.

الخطوات: افتح مجلد المثال، شغل الأمر، غير قيمة واحدة، ولاحظ النتيجة.

Mental model

Treat Plotly, react-plotly.js as one layer in a system. Identify the input, the transformation or rule, the output, and possible failures. Understanding that flow is more valuable than memorising syntax.

Checkpoint

Explain: What problem does this technology solve? What is its input? What output should it produce? What can fail?

Lesson 2 - Choosing a chart that matches the analytical question

What is it?

Start with the question: compare categories, show change over time, display distribution, or examine relationship. Choose the simplest chart that answers it.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
// Question: compare totals by month.
// A bar chart is clearer than a pie chart for this comparison.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Choosing a chart that matches the analytical question.

هذا الدرس يشرح: Choosing a chart that matches the analytical question. اكتب المثال بنفسك، حدد المدخلات والعملية والنتيجة، ثم غير قيمة واحدة لتفهم السلوك.

Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 3 - Traces, x/y data, layout, configuration, and rendering

What is it?

A trace contains data and mark type; layout controls titles, axes, legend, and spacing; config controls interactions such as mode bar and responsiveness.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
const data=[{type:'bar',x:['Jan','Feb'],y:[12,8]}];
const layout={title:'Monthly total'};
Plotly.newPlot('chart',data,layout,{responsive:true});
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Traces, x/y data, layout, configuration, and rendering.

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Lesson 4 - Bar, line, scatter, pie, histogram, and table trade-offs

What is it?

Bars compare categories, lines emphasise ordered change, scatterplots show relationships, histograms show distributions, pie charts suit very few parts, and tables preserve exact values.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
const bar={type:'bar',x:['A','B'],y:[4,8]};
const line={type:'scatter',mode:'lines+markers',x:['Jan','Feb'],y:[4,8]};
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Bar, line, scatter, pie, histogram, and table trade-offs.

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Quick check

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Lesson 5 - Labels, hover templates, legends, colours, and accessibility

What is it?

Use direct labels where practical, concise hover templates for detail, restrained colours, and a legend only when it removes ambiguity. Do not rely on colour alone.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
const trace={type:'bar',x:['Jan'],y:[12],
 text:['12 records'],textposition:'auto',
 hovertemplate:'%{x}: %{y}<extra></extra>'};
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Labels, hover templates, legends, colours, and accessibility.

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Lesson 6 - Sorting, top-N categories, percentages, and missing values

What is it?

Sort by analytical purpose, group the tail when justified, disclose top-N filtering, define percentage denominator, and represent missing values separately from zero.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
const rows=[{name:'A',value:20},{name:'B',value:8}];
rows.sort((a,b)=>b.value-a.value);
// Disclose if only top categories are shown.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Sorting, top-N categories, percentages, and missing values.

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Lesson 7 - Responsive sizing, margins, axes, and long category names

What is it?

Enable responsive sizing, reserve margins for labels, wrap or shorten long categories, set axis ranges deliberately, and test both narrow and wide screens.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
const layout={autosize:true,margin:{l:100,r:20,t:50,b:80},
 xaxis:{automargin:true},yaxis:{rangemode:'tozero'}};
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Responsive sizing, margins, axes, and long category names.

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Quick check

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Lesson 8 - Click and selection events for cross-filtering

What is it?

Plotly emits click and selection events containing points. Convert them into filter values, update application state, and clearly show or clear active cross-filters.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
chart.on('plotly_click', event => {
  const selected=event.points[0].x;
  applyFilter(selected);
});
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Click and selection events for cross-filtering.

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Quick check

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Lesson 9 - React Plotly components and immutable updates

What is it?

react-plotly.js receives data, layout, config, and handlers as props. Produce new objects when values change and separate transformation logic from UI rendering.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
<Plot data={traces} layout={layout} config={{responsive:true}}
 onClick={event => setFilter(event.points[0].x)} />
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates React Plotly components and immutable updates.

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Quick check

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Lesson 10 - Exporting SVG, PNG, and presentation-safe charts

What is it?

Vector SVG stays sharp; PNG needs sufficient dimensions and scale. Apply a high-contrast export theme and verify titles, legends, Arabic fonts, and margins.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
await Plotly.toImage(chart,{format:'png',width:1600,height:900,scale:2});
// Render and inspect labels, legend, and margins.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Exporting SVG, PNG, and presentation-safe charts.

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Quick check

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Lesson 11 - Testing chart transformations separately from rendering

What is it?

Unit-test functions that sort, aggregate, label, and build traces. Visual smoke tests then confirm Plotly renders the transformed structure as intended.

Why do we need it?

Without understanding this idea, code may appear to work while handling data incorrectly or failing on a different input. In Interactive Charts with Plotly, this concept helps create behaviour that is explicit, testable, and easier to change.

Syntax and example

```
test('sorts descending',()=>{
 expect(buildTrace(rows).x).toEqual(['A','B']);
});
// Test transformation before visual rendering.
```

What happens step by step

1. Read the example from the first line downward.
2. Identify the input or starting value.
3. Identify the operation, rule, or declaration.
4. Find the output or visible effect.
5. Predict the result before running it.
6. Run it and compare the real result with your prediction.

Try it yourself

Type the example instead of pasting it. Change one name and one synthetic value. Then introduce an empty or incorrect value and observe the result. Explain how this demonstrates Testing chart transformations separately from rendering.

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Quick check

Can you define the concept, point to it in the example, predict the output, and change the example without help? If not, repeat this lesson before continuing.

Lesson 7 - Guided mini-project

Project goal

Render a small interactive chart from synthetic totals.

Starting example

```
const data=[{type:'bar',x:['Jan','Feb'],y:[12,18]}];
Plotly.newPlot('chart',data,{title:'Synthetic totals'});
```

Read the code

Find imported tools or page elements. Identify the synthetic input. Find the function, event, route, or transformation that changes it. Finally, locate where the result is displayed, returned, saved, or packaged.

Build sequence

1. Run the untouched example.
2. Record the result.
3. Rename one unclear value.
4. Add a synthetic record.
5. Validate empty input.
6. Validate incorrect input.
7. Add a helpful error.
8. Rerun every case.
9. Compare with exercise-solution.
10. Explain every line.

Definition of done

Normal, empty, and invalid cases behave deliberately; no private data is used; and you can explain every line without reading the guide.

Lesson 8 - The real dashboard

Architecture connection

The application combines Plotly, react-plotly.js with the rest of the stack. The frontend gathers filters and files; the local API validates requests; the data layer transforms workbook rows; charts present aggregates; export tools create files; and the desktop layer packages the application for Windows.

Trace the upload feature

The user selects a workbook. The frontend sends it to the local API. The backend validates the file and sheets. The data layer creates safe tables. The API returns metadata. React updates state. Plotly displays results. Identify exactly where this guide's technology participates.

Privacy and quality

Do not place narrative, contact, or identifying fields in tutorials, logs, screenshots, tests, or exports. Use generated identifiers and synthetic totals. Validate at each boundary and collect only fields required for the calculation.

Architecture exercise

Draw five boxes: UI, API, Data Processing, Export, Desktop Packaging. Add arrows and label the data. Circle the box related to this guide and add two validation checks.

Lesson 9 - Debugging

Reliable debugging loop

1. Reproduce with the smallest input.
2. Read the first meaningful error.
3. Find the exact file and line.
4. Inspect value and type.
5. Compare expected and actual results.
6. Change one thing.
7. Rerun the same case.
8. Add a regression test.

Common beginner mistakes

- Wrong folder or command.
- Missing dependency or inactive environment.
- Misspelled file, variable, field, route, or import.
- Assuming input is always present.
- Mixing setup, processing, and display in one block.
- Hiding an exception rather than understanding it.
- Using real sensitive data in a test.

أخطاء شائعة: تشغيل الأمر من مجلد خاطئ، نسيان تثبيت التبعيات، أو تغيير أشياء كثيرة بمرة واحدة.

Error notebook

Record: command, expected result, actual error, root cause, smallest fix, and test added. This turns errors into reusable knowledge.

Lesson 10 - Exercises and answers

Exercises

1. Explain the technology in three sentences.
2. Recreate the example without copying.
3. Add one normal synthetic record.
4. Handle empty input.
5. Handle invalid input.
6. Improve unclear names.
7. Add or describe one test.
8. Locate the technology in the dashboard.
9. Record one error and root cause.
10. Add one mini-project feature.

تمرين: أضف شهرا أو قيمة آمنة جديدة، ثم اشرح ما حدث بكلماتك.

Answer guide

A strong solution is observable and explainable: the documented command works; output changes for the new record; empty and invalid cases have deliberate results; names communicate purpose; a test states an expected result; and the architecture explanation identifies the correct boundary.

Next step	Next: 10. Testing